

Let  $\lambda = (\lambda_t)_{t \geq 0}$  be a two-state Markov chain with  $Q$ -matrix  $Q$  taking values  $\lambda_l < \lambda_h$ , and let  $W$  be a standard Brownian motion.

The  $Q$ -matrix  $Q$  is given by

$$Q = \begin{pmatrix} -q_{lh} & q_{lh} \\ q_{hl} & -q_{hl} \end{pmatrix}$$

where  $q_{lh}$  is the rate of transition from  $\lambda_l$  to  $\lambda_h$  and  $q_{hl}$  is the rate of transition from  $\lambda_h$  to  $\lambda_l$ .

The initial distribution of  $\lambda$  is given by  $\mathbb{P}(\lambda_0 = \lambda_l) = p_0^l$  and  $\mathbb{P}(\lambda_0 = \lambda_h) = p_0^h = 1 - p_0^l$  and  $p_0 = (p_0^h, p_0^l)$ .

Let  $u$  be a control and consider the controlled SDE

$$dY_t^u = \lambda_t u_t dt + dW_t.$$

Defining  $p_t = \mathbb{P}(\lambda_t = \lambda_l | \mathcal{Y}_t^u)$ , we have

$$dp_t^u = (-q_{lh}p_t^u + q_{hl}(1 - p_t^u))dt + p_t^u(1 - p_t^u)(\lambda_h - \lambda_l)u_t dI_t^u, \quad p_0^u = p_0. \quad (.1)$$

## 0.1 Markovian lift

Hence, the Markovian lift is given by

$$\begin{aligned} dY_t^u &= (\lambda_h p_t^u + \lambda_l(1 - p_t^u))u_t dt + dI_t^u, & Y_0^u &= 0 \\ dp_t^u &= (-q_{lh}p_t^u + q_{hl}(1 - p_t^u))dt + p_t^u(1 - p_t^u)(\lambda_h - \lambda_l)u_t dI_t^u, & p_0^u &= p_0. \end{aligned}$$

The cost functional is given by

$$J(u) = \mathbb{E} \left[ \int_0^\infty e^{-\delta t} \left( \frac{1}{2} k (Y_t^u)^2 + \rho u_t^2 \right) dt \right].$$

On an infinite horizon, LQG problem, the HJB equation is given by

$$\begin{aligned} \sup_{u \in \mathcal{U}} \left\{ (\lambda_h p + \lambda_l(1 - p))u V_y + (-q_{lh}p + q_{hl}(1 - p))V_p \right. \\ \left. + \frac{1}{2} (V_{yy} + p^2(1 - p)^2(\lambda_h - \lambda_l)^2 u^2 V_{pp} + 2p(1 - p)(\lambda_h - \lambda_l)u V_{yp}) \right\} + \frac{1}{2} (ky^2 + \rho u^2) - \delta V = 0 \end{aligned}$$

and the minimizer is given by

$$u^* = - \frac{(\lambda_h p + \lambda_l(1 - p))V_y + p(1 - p)(\lambda_h - \lambda_l)V_{yp}}{p^2(1 - p)^2(\lambda_h - \lambda_l)^2 V_{pp} + \rho}.$$

Hence, the HJB equation is given by

$$\begin{aligned} - \frac{1}{2} \frac{\left( (\lambda_h p + \lambda_l(1 - p))V_y + p(1 - p)(\lambda_h - \lambda_l)V_{yp} \right)^2}{p^2(1 - p)^2(\lambda_h - \lambda_l)^2 V_{pp} + \rho} \\ + (-q_{lh}p + q_{hl}(1 - p))V_p + \frac{1}{2} V_{yy} + \frac{1}{2} ky^2 - \delta V = 0 \end{aligned}$$

and  $k, \rho, \delta > 0$  are model parameters.